Effect of Induced Molting on the Recurrence of a Previous *Salmonella enteritidis* Infection

PETER S. HOLT[1] and ROBERT E. PORTER, JR.[2]

USDA, Agricultural Research Service, Southeast Poultry Research Laboratory, 934 College Station Road, Athens, Georgia 30605

ABSTRACT Previous work in the authors' laboratory had shown that hens infected with *Salmonella enteritidis* (SE) during the feed removal phase of an induced molt shed significantly more SE and more readily transmitted SE to uninfected hens in adjacent cages when compared with unmolted hens. A study was conducted to examine the effect of induced molting on the recurrence and horizontal transmission of a previous SE infection. Hens aged 59 and 69 wk in Trials 1 and 2, respectively, were infected with SE and then molted 21 days later. In Trial 1, more molted hens were SE-culture-positive on Days 38 ($P \leq .005$) and 45 ($P \leq .005$) postinfection, and these hens shed more SE on these days ($P \leq .05$ and $P \leq .005$, respectively) than unmolted hens. Horizontal transmission of SE to previously uninfected but contact-exposed hens in adjacent cages was also higher in the molted group than the unmolted group on Days 38 ($P \leq .05$) and 45 ($P \leq .001$). Molted, contact-exposed hens also shed significantly more SE than unmolted hens. In Trial 2, the molted infected hens shed progressively more SE than the unmolted hens but the differences were not significant. However, more molted contact-exposed hens became SE-positive at Day 31 ($P \leq .05$) and 38 ($P \leq .005$) and also shed more SE on these days ($P \leq .05$ and $P \leq .01$, respectively) than the unmolted hens. Serum and intestinal antibody titers to SE were also examined in Trial 2. Molting appeared to exert no effect on the serum SE titers, but antibody titers in the alimentary tract were lower in the molted hens than the unmolted hens on Days 45 ($P \leq .005$) and 52 ($P \leq .05$). In Trial 1, three of eight molted directly infected hens and two of eight molted contact-exposed hens produced an SE-contaminated egg, but none of the unmolted hens produced any SE-contaminated eggs. In Trial 2, no SE-contaminated eggs were produced.

(*Key words*: stress, feed removal, *Salmonella enteritidis*, disease susceptibility, recrudescence)

1993 Poultry Science 72:2069–2078

INTRODUCTION

Induced molting is commonly used by the layer industry in the United States to stimulate multiple egg-laying cycles in hens (Bell, 1987). Although there are several methods to induce a molt, feed removal remains the primary means of achieving the egg-laying pause (Hembree *et al.*, 1978; Lee, 1982; Andrews *et al.*, 1987). However, recent research has shown hens infected with *Salmonella enteritidis* (SE) during the feed removal period had more severe intestinal infections than unmolted hens (Holt and Porter, 1992a), and this occurred in birds of different ages (Holt and Porter, 1992b). The molted infected hens had significantly more intestinal inflammation (Holt and Porter, 1992a),

Received for publication April 19, 1993.
Accepted for publication June 23, 1993.

[1]To whom correspondence should be addressed.
[2]Current address: Department of Pathobiology, School of Veterinary Medicine, Purdue University, West Lafayette, IN 47907.

shed significantly more SE from the alimentary tract (Holt and Porter, 1992a,b), and transmitted significantly more SE to uninfected, contact-exposed birds than unmolted hens (Holt and Porter, 1992b). As many flocks in the field may have been previously infected with SE and would be in various stages of clearing the infection, a group of studies were conducted to examine the effect of the induced molt feed removal procedure on hens that were infected with SE 3 wk previously. This study examined molting effects on intestinal shedding in the infected hens, the transmission of SE to uninfected hens in adjacent cages, the generation of antibody responses to SE over time, and the production of SE-contaminated eggs.

MATERIALS AND METHODS

Chickens

Birds used in the experiments were Single Comb White Leghorn hens retired from the laboratory's specific-pathogen-free production flock. The hens were housed in a totally light- and environment-controlled building and, until the commencement of the molt procedure, were maintained on a 16 h light (L):8 h dark (D) regimen (on at 0100 h). Unless otherwise specified, the hens were allowed *ad libitum* access to water and antibiotic-free pelleted feed (16.7% CP, 2,974 kcal ME/kg, 3.0% Ca, .44% P).

Experimental Protocol

A group of sixteen hens was divided equally into two main experimental groups, molt and control, and transferred into every other individual layer cage. Seven additional hens were placed in separate cages and were used as sentinel hens to follow the progression of the SE infection up to Day 21. These separate hens were not used further. Once transferred to their cages, all hens were allowed to acclimate for 1 wk and then infected with SE on Day 0. On Day 17, the empty cages adjacent to the 16 infected hens in the two experimental groups, along with their respective waterers, were disinfected. On Day 18, uninfected hens were transferred to these cages. These hens served as uninfected, contact-exposed hens to allow examination of the effect of molt on horizontal transmission of SE. There were therefore four test groups of eight hens each: molted infected, molted contact-exposed, unmolted infected, unmolted contact-exposed. Molt was induced on Day 21 as described below, and the directly infected and contact-exposed hens were followed for SE shedding for another 2 to 3 mo. The treatment and infection of hens was identical in both trials. However, several changes in sample handling were incorporated in Trial 2. These included tetrathionate enrichment of alimentary samples to increase the sensitivity of detection of intestinal SE shedding, sampling of egg pools rather than individual eggs to simplify the sampling procedure, and sampling of eggs both prior to and following molt to study the rate of production of contaminated eggs at various infection times.

Molting Procedure

Chickens were molted in each trial using a modification of a procedure described by Brake *et al.* (1982). On Day 14, the chickens were exposed to 8L:16D, which continued for 28 days at which time the light was increased to 10L:14D. The light was then increased by 2 h/2 wk thereafter until a final 16L:8D was achieved. Feed was removed from the molted group on Day 21, and the birds remained off feed for 14 days. Following the feed withdrawal period, the birds were fed a pullet grower ration (15% CP, 3,000 kcal ME/kg, .8% Ca, .32% P) for 14 days and then the layer breeder ration for the remainder of the experiment. Egg production ceased in all trials within 1 wk after withdrawal of feed.

Infection

Nalidixic acid-resistant SE Strain SE89-8312[3] is a phage Type 13 primary poultry isolate.[4] The nalidixic acid-resistant mutant used in this study was derived by A. Hinton

[3]Received from Arthur Hinton, Department of Botany and Microbiology, 129 Funchess Hall, Auburn University, Auburn, AL 36849.

[4]Originally obtained from the National Veterinary Services Laboratory, Ames, IA 50010.

and was maintained as a frozen stock at −20 C. Three days prior to infection, one of the stocks was thawed and subcultured for 2 consecutive days on nutrient agar[5] to eliminate any bacteria that were damaged from the freeze procedure. A 10-mL tube of tryptic soy broth was inoculated from the second plate. Birds to be infected received an oral dose by gavage of 1 mL of a 1:100 dilution of the overnight broth culture of the SE (5 to 10×10^6 bacteria as determined by dilution analysis) on Day 0. This dose was found to produce an intestinal infection in laying hens with detectable dissemination to organs in a small percentage of the birds. At the time of infection chickens of Trials 1 and 2 were 59- and 69-wk-old, respectively.

Intestinal Shedding of Salmonella enteritidis

The birds were assayed for intestinal shedding of SE using a modification of a procedure described by Holt and Porter (1992c). Sampling for SE clearance was conducted on Days 9 and 21 in the separate groups alluded to above. Once the molt was induced, sampling of the four treatment groups was performed weekly beginning on Day 3 of molt (Day 24 of infection). Eighteen hours prior to each sample collection, feed was removed from the chickens and on the test day each birds received an intraperitoneal injection of .5 mL of 5% pilocarpine.[6] This drug induces hypersecretion of the intestine, and the resulting secretions provide an excellent sample for evaluating intestinal shedding of SE (Holt and Porter, 1992c). Repeated weekly sampling had no apparent effect on the course of an intestinal SE infection (data not shown). The alimentary secretions were collected over the next hour in 23 × 28 cm food-grade quality styrofoam trays[7] placed under each individual layer cage. The secretions were serially diluted (10-fold) in PBS, and 100 μL of each dilution was spread onto brilliant green agar containing 20 μg novobiocin/mL and 20 μg nalidixic acid/mL (BGNN). The plates were incubated overnight at 37 C, and bacterial counts were made. In Trial 2, 1 mL of alimentary secretion was also transferred to 9 mL tetrathionate enrichment broth, and these samples were incubated overnight at 37 C. One hundred microliters of the tetrathionate samples were streaked onto BGNN. The plates were incubated at 37 C and then were examined for the presence of SE colonies. The minimum threshold of SE detection in Trial 1 was 10 SE/mL of alimentary secretions, and samples with no detectable SE were considered negative for recoverable SE but for enumeration purposes were presumed to contain 9 SE/mL. In Trial 2, because of the tetrathionate enrichment, the level of SE detection was 1 SE/mL alimentary secretion. Samples that had no growth on the primary culture plates and the tetrathionate enrichment were considered negative for recoverable SE and assigned a value of 0 SE/mL. Samples that had no growth on the primary culture plates but were culture-positive following tetrathionate enrichment were considered positive for recoverable SE and assigned 9 SE/mL.

Antibody Responses

Molted and unmolted hens were bled weekly with heparinized syringes at the same time the hens were sampled for intestinal shedding. The blood and alimentary secretion samples were centrifuged at $1{,}700 \times g$, and the supernatants were frozen at −20 C until time of assay. The plasma and alimentary secretions were assayed by ELISA using SE flagella, purified by the method of Ibrahim *et al.* (1985), as the solid-phase detection antigen. One hundred microliters of the flagellar antigen at 1 μg/mL and 2 μg/mL in carbonate buffer pH 9.2 for plasma and alimentary secretion assays, respectively, were dried onto individual wells of Immulon II plastic trays[8] overnight at 37 C. The trays were then sealed and stored at 4 C until time of assay. The wells were then washed for 2 min with PT buffer (phosphate-buffered saline, pH 7.2, plus 1% Tween 20) and then a further 25 min with

[5]All bacteriological media were obtained from Difco Laboratories, Detroit, MI 48232.

[6]All chemicals and antibiotics were obtained from Sigma Chemical Co., St. Louis, MO 63718-9916.

[7]Genpack Corp., Glens Falls, NY 12801.

[8]Dynatech Laboratories, Inc., Chantilly, VA 22021.

PT buffer containing .1% bovine serum albumin (PTB) to block nonspecific binding. The wells were washed three times with PT buffer, 2 min per wash, and then 100 μL plasma (diluted 1:250 in PTB) or 100 μL alimentary secretion (diluted 1:10 in PTB) were added to duplicate wells. Seven plasma or alimentary secretion samples per group were assayed for each sampling day. The trays were incubated at 25 C for 90 min and then were washed three times with PT buffer.

Monoclonal antibodies developed in the authors' laboratory specific for chicken IgG and IgA were used in the plasma and alimentary secretion assays, respectively. These antibodies were diluted 1:40 and 1:5 in PTB for the anti-IgG and anti-IgA, respectively, and 100 μL were added to their respective trays. The trays were incubated at 25 C a further 90 min and then were washed three times with PT buffer. One hundred microliters of 1:1000 dilution of alkaline phosphatase-labeled goat anti-mouse IgG[9] was added to each well, and the trays were incubated for 60 min at 25 C. The trays were washed three times with PT, and then 100 μL of the *p*-nitrophenol phosphate substrate, 1 mg/mL, was added to each well. The trays were incubated in the dark at 25 C for 20 min and 90 min for the IgG and IgA samples, respectively, at which time 20 μL of 3*M* NaOH was added to each well to stop the reaction.

The color reaction was read at 405 nm using a MS2/MCC 340 ELISA reader.[10] The readings were converted to a positive to negative (P:N) ratio by dividing the average ELISA reading of the sample by the average ELISA reading of six wells containing plasma or alimentary secretions from uninfected White Leghorn hens. These control samples were diluted and treated with primary and secondary antibody as described above. The P:N ratios were converted to natural logarithms, and titers were determined as described by Briggs *et al.* (1986) using the equation $[\ln(P{:}N) \times 1.427] + 5.575$ for plasma and $[\ln(P{:}N) \times 1.348] + 2.628$ for alimentary secretions. These equations were derived from 100 plasma or secretion samples from individual infected White Leghorn hens. The titers of these original samples were determined by serial dilution, and the ELISA reading at 1:250 dilution (plasma) and 1:10 dilution (alimentary secretion) was also obtained. These readings were plotted as a function of titer, and the respective equation was derived from each of the line plots.

Egg Culture

In Trial 1, eggs were collected from all four groups from Day 72 to 129. A total of 435 eggs were collected and cultured. In Trial 2, eggs were collected from the infected hens from Day 3 to 14 (69 eggs) and then on Days 55 to 87 once lay had restarted (186 eggs). The eggs in both trials sat at room temperature for 1 wk, and the shells were disinfected by emersing each egg in 2% tincture of iodine just prior to sampling. In Trial 1, individual egg contents were emulsified in 40 mL tryptone soya broth supplemented with 35 mg/L ferrous sulfate, and the cultures were incubated at 37 C for 72 h. One hundred microliters of each culture was spread onto BGNN, and, following an 18-h incubation at 37 C, the plates were examined for colonies indicative of SE. In Trial 2, the egg contents were pooled weekly by hen. Twenty milliliters of each pool was added to 160 mL of tryptone soya broth supplemented with 35 mg/L ferrous sulfate, and the cultures were incubated at 37 C for 18 h. As in Trial 1, one hundred microliters of each culture was spread onto BGNN, and, following an 18-h incubation at 37 C, the plates were examined for colonies indicative of SE.

Statistical Analysis

The numbers of SE/mL shed in each treatment group (molted versus unmolted) were transformed to $\log_{10}$, and then means were calculated. The means for the plasma and alimentary secretion titers were also calculated. Significant differences between treatment groups of the mean $\log_{10}$ SE/mL, the percentage of hens positive for SE, and the plasma and alimentary secretion titers were calculated using the pooled-variance *t* test (Shott, 1990).

---

[9]Calbiochem Corp., La Jolla, CA 92037.

[10]Flow/ICN Laboratories, Costa Mesa, CA 92626.

RESULTS

Intestinal Shedding of Directly Infected Hens

In Trial 1, 71% of the hens at Day 9 were shedding SE (Figure 1A) for an average 2.2 logs SE per milliliter of alimentary secretions (Figure 2A). This number dropped to 14% (1.5 logs SE per milliliter) by Day 21, the 1st day of feed removal. By Day 24, SE could no longer be detected in the unmolted hens, and this continued until the trial was terminated at Day 127. In the molted hens, however, 25% of the hens were shedding SE on Days 24 and 31 for an average of 1.86 and 1.87 logs SE per milliliter, respectively. The percentages increased to 62.5% on Days 38 ($P \leq .005$) and 45 ($P \leq .005$), 3 and 10 days, respectively, after the molted hens resumed feeding. These hens shed an average of 2.04 logs SE per milliliter ($P \leq .05$) and 2.39 logs SE per milliliter ($P \leq .005$), respectively, on these 2 days. The percentage of molted hens that were shedding SE gradually decreased, but there were still some hens (25%) chronically shedding SE at Day 127 when the trial was terminated. One hen in this trial, Hen 79, was notable in her shedding of SE, and Figure 3 graphically depicts the amount of SE shed by this hen over the course of the trial. Hen 79 was infected on

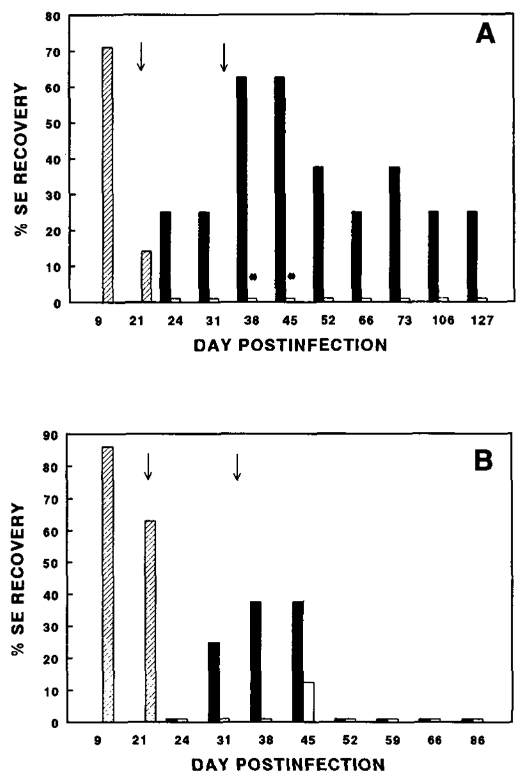


FIGURE 1. Effects of the induced molt feed withdrawal procedure on *Salmonella enteritidis* (SE) shed rate in previously infected hens in Trial 1 (A) and Trial 2 (B). Results represent the percentage of hens that were SE-positive on the days sampled; Lined bar = sentinel group of hens to follow SE shedding (n = 7 hens), Filled bar = molted hens, Open bar = unmolted hens (n = 8 hens each per molted and unmolted groups); Arrows = Day 21 feed removed and Day 35 feed resumed; * = significantly different from unmolted group, $P \leq .05$.

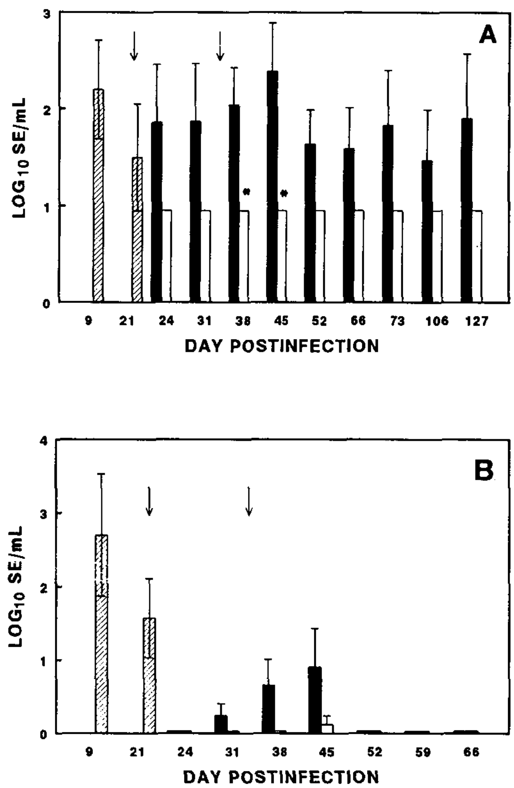


FIGURE 2. Effects of the induced molt feed withdrawal procedure on intestinal *Salmonella enteritidis* (SE) numbers in previously infected hens in Trial 1 (A) and Trial 2 (B). Results represent the $\log_{10}$ number SE per milliliter of alimentary secretions on the days sampled; Lined bar = sentinel group of hens to follow SE shedding, Filled bar = molted hens, Open bar = unmolted hens (n = 8 hens each per molted and unmolted groups); Arrows = Day 21 feed removed and Day 35 feed resumed; * = significantly different from unmolted group, $P \leq .05$.

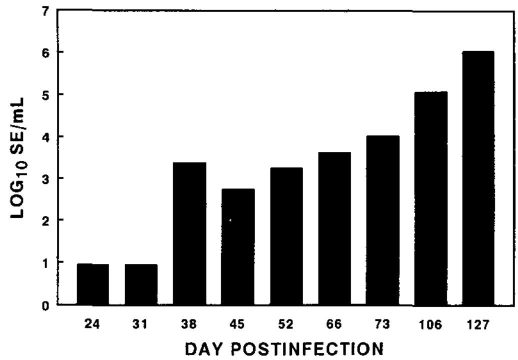


FIGURE 3. Shedding of *Salmonella enteritidis* (SE) from the alimentary tract of Hen 79. Results represent the $\log_{10}$ number SE per milliliter of alimentary secretion from Hen 79 on the days sampled.

Day 0 and molted on Day 21. Hen 79 was not sampled until Day 24 and no SE was detected at this time nor at Day 31. However, by Day 38, Hen 79 was shedding over 3 logs SE and this gradually increased to over 6 logs SE by Day 127. This hen showed no overt signs of disease except for a slight sour body odor and gave no indication that she was shedding large numbers of the SE organism.

In Trial 2, 86% of the tested hens were SE culture-positive on Day 9 (Figure 1B) and shed 2.7 logs of SE per milliliter of alimentary secretion (Figure 2B). On Day 21, 63% of the hens were SE culture-positive, shedding an average 1.6 logs SE. By Day 24, the percentage of hens shedding SE dropped to 0 in both groups. On Day 31, the percentage of SE culture-positive molted hens had increased to 25% and increased further to 37.5% by Day 38, but the unmolted hens continued to be SE culture-negative. The hens were inadvertently deprived of feed for 36 to 48 h during the sampling at this time period, and on Day 45 one of the unmolted hens (12.5%) became SE culture-positive compared with three of eight (37.5%) of the molted hens. No SE could be detected in either group of hens from Day 52 to the conclusion of the trial at Day 86.

Intestinal Shedding of Contact-Exposed Hens

The molting procedure also affected the transmission of SE and the degree of intestinal infection by SE in previously uninfected, contact-exposed birds located in adjacent cages. The hens in both molted and unmolted treatment groups in Trial 1 were infected to a similar degree at Day 24 postinfection (Figures 4A and 5A), but thereafter the percentage of molted contact-exposed hens that became SE-positive increased such that 50% ($P \leq .05$) and 75% ($P \leq .001$) of the molted hens were shedding SE on Days 38 and 45, respectively, compared with 0% of the unmolted hens. Alimentary secretions from these birds at Days 38 and 45 contained more SE per milliliter (1.5 to 2 logs) than the unmolted group (Figure 5A, $P \leq .05$ and $P \leq .001$, respectively). The percentage of SE-positive molted hens decreased thereafter, as did the amount of

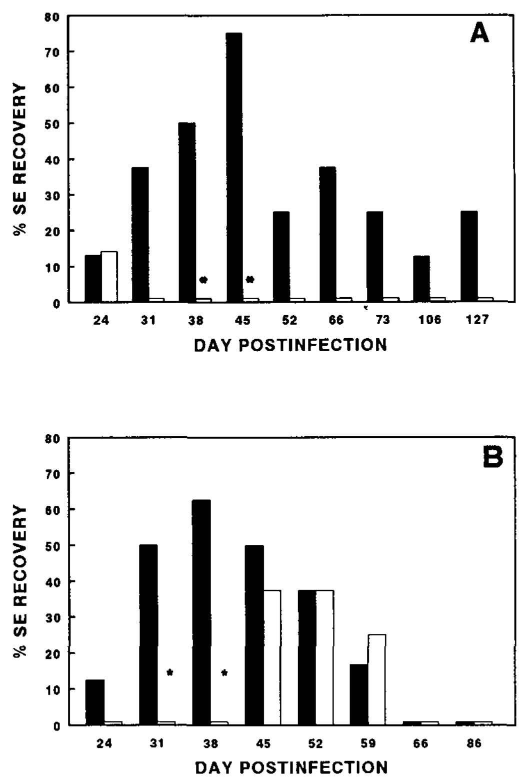
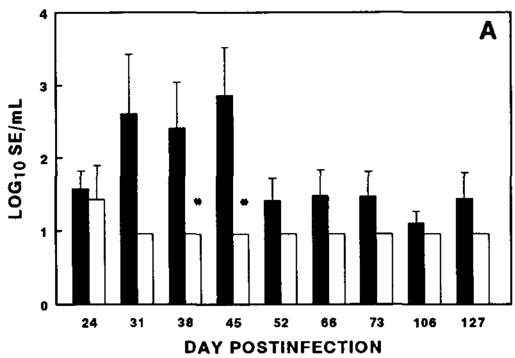


FIGURE 4. Effects of the induced molt feed removal procedure on *Salmonella enteritidis* (SE) shed rate of previously uninfected, contact-exposed hens in Trial 1 (A) and Trial 2 (B). Results represent the percentage of hens that were SE-positive on the days sampled (n = 8 hens per treatment group); Filled bar = molted hens, Open bar = unmolted hens; * = significantly different from unmolted group, $P \leq .05$.

SE shed, but there were still a certain percentage of SE-positive hens at Day 127 when the trial was terminated.

In Trial 2 (Figure 4B), more ($P \leq .05$) molted contact-exposed hens became SE-positive at Day 31, and these hens shed 2.6 logs (5B, $P \leq .05$) more SE than their unmolted counterparts. By Day 38, this percentage had increased to 63% ($P \leq .005$), and the hens were shedding 3.7 logs per milliliter of alimentary secretion ($P \leq .01$). On Day 45, 1 wk after the hens were accidently left off feed 36 to 48 h, 50% of the molted, contact-exposed hens were SE-positive, and 38% of the unmolted hens were also shedding SE. The SE shedding was equivalent in both groups of birds for the remainder of the trial.



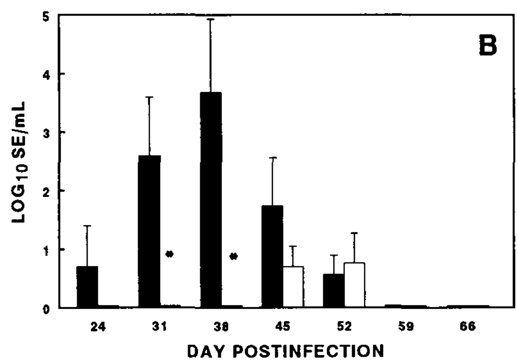


FIGURE 5. Effects of the induced molt feed removal procedure on intestinal *Salmonella enteritidis* (SE) numbers of previously uninfected, contact-exposed hens in Trial 1 (A) and Trial 2 (B). Results represent the $\log_{10}$ number SE per milliliter of alimentary secretions on the days sampled (n = 8 hens per treatment group); Filled bar = molted hens, Open bar = unmolted hens; * = significantly different from unmolted group, $P \leq .05$.

Antibody Titers to Salmonella enteritidis *in Molted and Unmolted Hens*

In Trial 2, antibody titers to SE in the plasma rose from 8.32 on Day 9 to 9.61 on Day 21 (Figure 6A). On Day 24, 3 days into the feed removal, the titers in the molted hens was 7.3 compared with 7.82 in the unmolted hens. The plasma titers remained equivalent for the remainder of the trial. Antibody titers in the alimentary tract followed a different trend (Figure 6B). On Day 9, the alimentary secretion titer was 4.91 and on Day 21 the titer had increased to 6.2. On Day 24, the titer in the molted hens was 5.86 compared with 4.95 in the unmolted birds. The titers were equivalent for both groups of hens on Days 31 and 38. However, on Days 45 and 52, the titers were higher in the unmolted hens ($P \leq .005$ and $P \leq .05$, respectively).

Recovery of Salmonella enteritidis *from Eggs*

In Trial 1, SE was recovered from five eggs, each from a different hen (Table 1). All of the isolations were from molted hens; three isolations from infected hens and two isolations from molted, contact-exposed hens. Hen 79 was one of the hens that produced an SE-contaminated egg. In Trial 2, no SE-contaminated eggs were detected.

DISCUSSION

The current authors had shown previously that inducing a molt in older hens using a 14-day feed removal protocol significantly exacerbated an intestinal infection by SE (Holt and Porter, 1992a,b). These infections in the previous studies were initiated concurrently with the feed removal. Dietary restriction has been shown previously to increase susceptibility to disease (Formal *et al.*, 1958; McGuire *et al.*, 1968; Chandra and Newberne, 1977), and Miller and Bohnhoff (1962) showed that the timing of a feed removal period relative to an oral SE infection would also affect the infectivity of SE for mice. In the current study, hens were subjected to the feed withdrawal 21 days after infection, and this treatment caused a recurrence of

TABLE 1. Recovery of *Salmonella enteritidis* (SE) from eggs layed by molted and unmolted hens

Trial	Treatment	SE isolations/total eggs cultured (percentage recovery)[1]	
		Infected	Contact-exposed
1	Unmolted	0/105 (0%)	0/53 (0%)
1	Molted	3/153 (1.9%)	2/124 (1.6%)
2	Unmolted	0/58 (0%)	0/59 (0%)
2	Molted	0/59 (0%)	0/34 (0%)

[1]Results represent the number of SE-positive eggs/total eggs cultured, with the percentage recovery in parentheses. In Trial 2, 69 eggs cultured from Days 3 to 14 were negative for SE.

the SE infection. This was especially evident in Trial 1 with significantly higher shed rates in molted than unmolted hens on Days 38 and 45. Similar, though nonsignificant, effects were observed in Trial 2. Stress has been shown previously to cause the reactivation of a previous infection (Soave, 1964), and the exacerbation of infection in molted hens in the current study may be a result of molt stress. Interestingly, although induced molting exacerbates an SE infection whether the hens are deprived of feed concurrently with the challenge or 21 days after the challenge, the manifestation of these effects occur differently relative to the feed removal. Hens challenged with SE during the feed removal period generally shed significantly more SE than unmolted hens during the feed removal period (Holt and Porter, 1992a,b). Hens receiving the SE 3 wk prior to feed withdrawal, however, shed significantly higher numbers of SE after the feeding was resumed than unmolted hens (Figure 2). The different effects observed in the latter case may be a reflection of the low numbers of SE found in the infected hens at the time of feed removal, which then required time to increase the intestinal SE to significant levels. Other factors such as altered intestinal immunity (Figure 6B) may also play a role and need to be examined further.

Hens infected with SE concurrently during the feed removal portion of the induced molt have been previously shown to horizontally transmit SE to uninfected, contact-exposed hens in adjacent cages at a significantly higher rate than in unmolted hens (Holt and Porter, 1992b). Similar results were observed in the

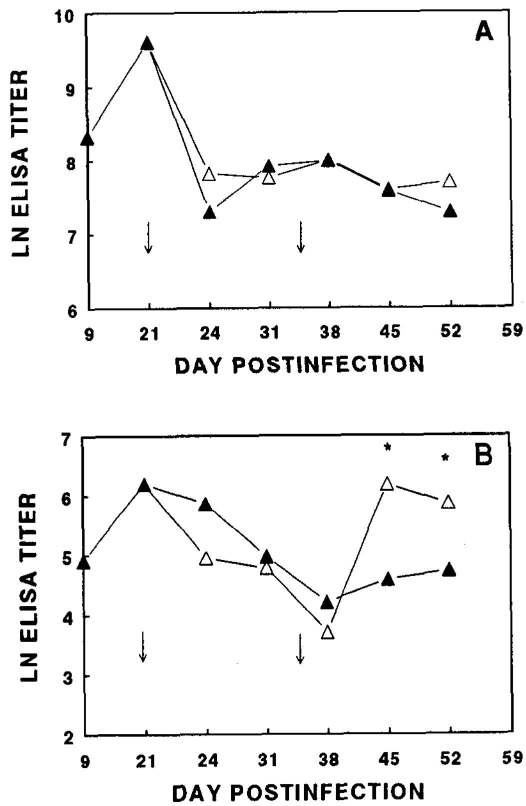


FIGURE 6. Effects of the induced molt feed removal procedure on anti-*Salmonella enteritidis* (SE) antibody titers in serum (A) and alimentary secretions (B). (n = samples from 7 hens per treatment group on the days sampled); Filled triangle = molted hens, Open triangle = unmolted hens; Arrows = Day 21 feed removed and Day 35 feed resumed; * = significantly different from unmolted group, P ≤ .05.

current study (Figure 3A and B), and the kinetics of this transmission followed a pattern similar to that observed for the recrudescence of the primary infection. This increased transmissibility may be due to the increased numbers of SE shed by the molted infected hens (Figure 2) and the observed increased susceptibility of molted hens to an SE infection (Holt, 1993).

The progression of the infection of Hen 79 (Figure 3) provides evidence of the subtle and insidious nature of SE. This hen had been infected with SE and had cleared the organism to below detectable levels. However, once molted, Hen 79 began shedding SE at progressively higher levels over time. Besides producing an SE-contaminated egg, this hen could also be a major source of transmission of SE to other hens and also be a source of contamination of the layer facility environment that could affect future flocks (Opitz, 1992). The factors that induce a SE culture-negative hen to begin shedding large numbers of SE remain to be determined.

Previous studies had shown that cell-mediated immunity was significantly depressed by the molt procedure whereas serum humoral immunity was unaffected (Holt, 1992). In Trial 2 of the current study, serum humoral immunity to SE was unaffected, but the local immune response of the alimentary tract was significantly lower in the molted hens on Days 45 and 52 than unmolted hens (Figure 6). The systemic immune system and the local intestinal immune system appear to be compartmentalized (Mestecky, 1987; Hone and Hackett, 1989), and their regulation also appears to be different (Challacombe and Tomasi, 1980; Hone and Hackett, 1989). Different responses in the alimentary secretions and sera in molted hens in the current study may be a reflection of this compartmentalization.

Salmonella enteritidis became a problem for the layer industry because of the presence of the organism in intact table eggs (St. Louis *et al.*, 1988; O'Brien, 1990). Previous studies have shown that although egg SE contamination can be found readily, the percentage of positive eggs is low (Gast and Beard, 1990, 1992; Shivaprasad *et al.*, 1990). The dosage of SE used in the current study was purposely chosen to be lower than that normally used for infecting hens (Gast and Beard, 1990, 1992), and this dose has been found to disseminate extraintestinally in a low percentage of cases (data not shown). Unmolted hens did not produce any SE-contaminated eggs in either trial whereas in Trial 1, SE was recovered from five different molted hens. No SE was recovered in Trial 2. A previous study reported the sporadic recovery of SE from eggs of molted SE-infected hens (Holt and Porter, 1992a), but overall the SE egg contamination rate was extremely low in both studies. The effect of induced molting on the production of SE-contaminated eggs therefore remains unclear. However, the large amount of SE that could be shed into the house environment during a molt, as alluded to above, may pose a greater problem for the producer and for future flocks that would occupy that house.

ACKNOWLEDGMENTS

The authors wish to thank Amy Beyer and Lisa Higginbotham for their excellent technical assistance during the study.

REFERENCES

Andrews, D. K., W. D. Berry, and J. Brake, 1987. Effect of lighting program and nutrition on reproductive performance of molted Single Comb White Leghorn hens. Poultry Sci. 66: 1298–1305.

Bell, D., 1987. Is molting still a viable replacement alternative? Poult. Trib. 95:32–35.

Brake, J., G. W. Morgan, M. Baler, and P. Thaxton, 1982. Physiological changes in caged layers during a forced molt. 4. Leucocyte and packed cell volume. Poultry Sci. 61:790–795.

Briggs, D. J., C. E. Whitfill, J. K. Skeeles, J. D. Story, and K. D. Reed, 1986. Application of the positive/negative ratio method of analysis to quantitate antibody responses to infectious bursal disease virus using a commercially available ELISA. Avian Dis. 30:216–218.

Challacombe, S. J., and T. B. Tomasi, 1980. Systemic tolerance and secretory immunity after oral immunization. J. Exp. Med. 152:1459–1472.

Chandra, R. K., and P. M. Newberne, 1977. Pages 127–180 *in*: Nutrition, Immunity, and Infection Mechanisms of Interactions. Plenum Press, New York, NY.

Formal, S. B., G. J. Dammin, E. H. LaBrec, and H. Schneider, 1958. Experimental Shigella infections: characteristics of a fatal infection produced in guinea pigs. J. Bacteriol. 75:604–610.

Gast, R. K., and C. W. Beard, 1990. Production of Salmonella enteritidis-contaminated eggs by experimentally infected hens. Avian Dis. 34: 438–446.

Gast, R. K., and C. W. Beard, 1992. Detection and enumeration of *Salmonella enteritidis* in fresh and stored eggs laid by experimentally infected hens. J. Food Prot. 55:152–156.

Hembree, D. J., A. W. Adams, and J. V. Craig, 1978. Effects of light restriction and amino acid supplementation on performance and agonistic behavior of forced-molted hens. Poultry Sci. 57: 1143–1144.

Holt, P. S., 1992. Effects of induced molting on immune responses of hens. Br. Poult. Sci. 33: 165–175.

Holt, P. S., 1993. Effect of induced molting on the susceptibility of White Leghorn hens to a Salmonella enteritidis infection. Avian Dis. 37: 412–417.

Holt, P. S., and R. E. Porter, Jr., 1992a. Microbiological and histopathological effects of an induced molt fasting procedure on a Salmonella enteritidis infection in chickens. Avian Dis. 36:610–618.

Holt, P. S., and R. E. Porter, Jr., 1992b. Effect of induced molting on the course of infection and transmission of *Salmonella enteritidis* in White Leghorn hens of different ages. Poultry Sci. 71: 1842–1848.

Holt, P. S., and R. E. Porter, Jr., 1992c. Use of pilocarpine-induced alimentary secretions to measure intestinal shedding of Salmonella enteritidis in chickens. Avian Dis. 36:54–58.

Hone, D., and J. Hackett, 1989. Vaccination against enteric bacterial diseases. Rev. Infect. Dis. 11: 853–877.

Ibrahim, G. F., G. H. Fleet, M. J. Lyons, and R. A. Walker, 1985. Method for the isolation of highly purified Salmonella flagellins. Infect. Immun. 22:1040–1044.

Lee, K., 1982. Effects of forced molt period on postmolt performance of leghorn hens. Poultry Sci. 61:1594–1598.

McGuire, E. A., V. R. Young, P. M. Newberne, and B. J. Payne, 1968. Effects of *Salmonella typhimurium* infection in rats fed varying protein intakes. Arch. Pathol. 86:60–68.

Mestecky, J., 1987. The common mucosal immune system and current strategies for induction of immune responses in external secretions. J. Clin. Immunol. 7:265–276.

Miller, C. P., and M. Bohnhoff, 1962. A study of experimental *Salmonella* infection in the mouse. J. Infect. Dis. 111:107–116.

O'Brien, J.D.P., 1990. Aspects of *Salmonella enteritidis* control in poultry. World's Poult. Sci. J. 46: 119–124.

Opitz, H. M., 1992. Progress being made in *Salmonella enteritidis* reduction on the farm. Poult. Dig. 51: 16–22.

St. Louis, M. E., D. L. Morse, M. E. Potter, T. M. DeMelfi, J. J. Guzewich, R. V. Tauxe, and P. A. Blake, 1988. The emergence of Grade A eggs as a major source of *Salmonella enteritidis* infections: new implications for the control of salmonellosis. J. Am. Med. Assoc. 259:2103–2107.

Shivaprasad, H. L., J. F. Timoney, S. Morales, B. Lucio, and R. C. Baker, 1990. Pathogenesis of Salmonella enteritidis infection in laying chickens. I. Studies on egg transmission, clinical signs, fecal shedding, and serologic responses. Avian Dis. 34:548–557.

Soave, O. A., 1964. Reactivation of rabies virus in a guinea pig due to the stress of crowding. Am. J. Vet. Res. 25:268–269.

Shott, S., 1990. Hypothesis testing. Pages 105–143 *in*: Statistics for Health Professionals. W. B. Saunders Company. Philadelphia, PA.